rather than individually. Rate-limiting attempts to ensure that large floods of requests are rejected before the system is completely overwhelmed and ceases functioning entirely.

> **DEFINITION *Throttling*** is a process by which a client's requests are slowed down without disconnecting the client completely. Throttling can be achieved either by queueing requests for later processing, or else by returning a status code telling the client to slow down.

The most important aspect of rate-limiting is that it should use fewer resources than would be used if the request was processed normally. For this reason, rate-limiting is often performed in highly optimized code running in an off-the-shelf load balancer, reverse proxy or API gateway that can sit in front of your API to protect it from DoS attacks rather than having to add this code to each API.

In the next chapter we will get our hands dirty with a real API and apply some of the techniques we have discussed in this chapter.

EXERCISES

6. Which of the STRIDE threats does rate-limiting protect against?

a) Spoofing

b) Tampering

c) Repudiation